

Original Article

The water extract of Liuwei dihuang possesses multi-protective properties on neurons and muscle tissue against deficiency of survival motor neuron protein



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ABSTRACT

Background: Deficiency of survival motor neuron (SMN) protein, which is encoded by the *SMN1* and *SMN2* genes, induces widespread splicing defects mainly in spinal motor neurons, and leads to spinal muscular atrophy (SMA). Currently, there is no effective treatment for SMA. Liuwei dihuang (LWDH), a traditional Chinese herbal formula, possesses multiple therapeutic benefits against various diseases via modulation of the nervous, immune and endocrine systems. Previously, we demonstrated water extract of LWDH (LWDH-WE) protects dopaminergic neurons and improves motor activity in models of Parkinson's disease.

Purpose: This study aimed to investigate the potential protection of LWDH-WE on SMN deficiency-induced neurodegeneration and muscle weakness.

Study design: The effects of LWDH-WE on SMN deficiency-induced neurotoxicity and muscle atrophy were examined by using SMN-deficient NSC34 motor neuron-like cells and SMA-like mice, respectively.

Methods: Inducible *SMN*-knockdown NSC34 motor neuron-like cells were used to mimic SMN-deficient condition. Doxycycline (1 µg/ml) was used to induce SMN deficiency in stable NSC34 cell line carrying *SMN*-specific shRNA. SMAΔ7 mice were used as a severe type of SMA mouse model. Cell viability was measured by 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl-tetrazolium bromide (MTT) and lactate dehydrogenase (LDH) assays. Apoptotic cells and neurite length were observed by inverted microscope. Protein expressions were examined by western blots. Muscle strength of animals was evaluated by hind-limb suspension test.

Results: LWDH-WE significantly increased SMN protein level, mitochondrial membrane potential and cell viability of SMN-deficient NSC34 cells. LWDH-WE attenuated SMN deficiency-induced down-regulation of B-cell lymphoma-2 (Bcl-2) and up-regulation of cytosolic cytochrome c and cleaved caspase-3. Moreover, LWDH-WE prevented SMN deficiency-induced inhibition of neurite outgrowth and activation of Ras homolog gene family, member A (RhoA)/ Rho-associated protein kinase (ROCK2)/ phospho-LIM kinase (p-LIMK)/ phospho-cofilin (p-cofilin) pathway. Furthermore, in SMA-like mice, LWDH-WE improved muscle strength and body weight accompanied with up-regulation of SMN protein in spinal cord, brain, and *gastrocnemius* muscle tissues.

Conclusion: The present study demonstrated that LWDH-WE protects motor neurons against SMN deficiency-induced neurodegeneration, and it also improves the muscle strength of SMA-like mice, suggesting the potential benefits of LWDH-WE as a complementary prescription for SMN deficiency-related diseases.

Introduction

Survival motor neuron protein (SMN) is a ubiquitous and

indispensable protein that is essential for a series of basic cellular processes, including transcription, pre-mRNA splicing, small nuclear ribonucleoproteins (snRNPs) biogenesis (Burghes and Beattie, 2009;

Abbreviations: LWDH-WE, Liuwei dihuang water extract; SMN, survival motor neuron; MNDs, motor neuron diseases; Bcl-2, B-cell lymphoma 2; RhoA, Ras homolog gene family, member A; ROCK, Rho-associated protein kinase; LIMK, LIM kinase; SMA, spinal muscular atrophy

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Coady and Lorson, 2011). SMN also plays a critical role in survival especially in motor neurons. It is suggested that neuron-specific loss of SMN leads to motor neurons degeneration through an apoptotic mechanism (Gallotta et al., 2016). Besides, SMN is able to modulate axonal mRNA transport and plays a critical role in maintaining the integrity of spinal and neuromuscular circuitry (Lin et al., 2016; Ling et al., 2010; Rathod et al., 2012).

Progressive motor neuronal degeneration is an identified characteristic of motor neuron diseases (MNDs) that can further result in muscle atrophy and weakness (Rossi et al., 2012; Saxena and Caroni, 2011). Spinal muscular atrophy (SMA) is one of the MNDs, resulting from *SMN1* gene deletion or mutation, leads to pathological decrease of functional SMN protein and loss of motor neurons (Lefebvre et al., 1995). It is known that actin cytoskeleton has an important role in neurite outgrowth, and SMN in motor neuron growth cones is responsible for actin mRNA localization (Rossoll et al., 2003). Besides SMN, small GTPase RhoA and its major downstream effector Rho-associated protein kinase (ROCK) are also known as actin cytoskeleton regulators during neuronal development, which both contribute to the pathology of MNDs (Chong et al., 2017). RhoA/ROCK activation can promote actin cytoskeletal collapse and induce neurite retraction (Govek et al., 2005; Lingor et al., 2007). Moreover, the increased activity of RhoA in the spinal cord of a SMA mouse model suggests the possibility of RhoA activation caused by low level of SMN (Bowerman et al., 2010). Additionally, pharmacological inhibition of RhoA/ROCK pathway is found to improve motor neuronal outgrowth and neuromuscular junction maturation in SMA cellular and mouse models (Bowerman et al., 2010; Coque et al., 2014), suggesting RhoA/ROCK inhibition as a potential therapeutic target in motor neuronal protection against SMN deficiency.

The Chinese herbal formula Liuwei dihuang (LWDH), composed of dihuang (*Rehmannia glutinosa*), shanyao (*Dioscorea opposita*), shanzhuyu (*Cornus officinalis*), zexie (*Alisma orientalis*), hoelen (*Poria cocos*) and mudanpi (*Paeonia suffruticosa*), has been used for replenishing “Yin” of the kidney traditionally. In the theory of traditional Chinese medicine, the “kidney” is described as the fundamental system for reproduction, development and performance. Accordingly, “kidney yin deficiency” reflects many kinds of imbalance in physiological functions of the body including the nervous system (Zhou et al., 2016). Besides, the pathogenesis of MNDs are often accompanied by the syndrome of kidney yin deficiency stages, such as weakness and soreness of the waist and knees, vertigo, tinnitus, deafness, night sweats and emissions. Our previous study suggests water extract of LWDH (LWDH-WE) as a potential neuroprotective prescription that can protect dopaminergic neurons against Parkinson's toxin via anti-oxidative and anti-apoptotic abilities, and can improve motor activity of Parkinson's disease mice (Tseng et al., 2014). We also revealed the neuroprotective effects of two active components of LWDH, paeonol and loganin (Tseng et al., 2016, 2012), and first mentioned the potential muscle-enhancing effect of loganin in SMA via targeting protein synthesis Akt/mTOR pathway (Tseng et al., 2016). However, the potential of LWDH on SMA treatment has not been studied. Therefore, the present study aimed to investigate the protective effects and mechanisms of LWDH-WE in experimental models of SMA.

Materials and methods

Materials

Doxycycline, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), and Hoechst33342 were obtained from Sigma-Aldrich (St. Louis, MO, USA). 5,5',6,6'-tetrachloro-1,3',3,3'-tetraethylbenzimidazolylcarbocyanine iodide (JC-1), Dulbecco's modified Eagle's medium (DMEM) and all culture medium supplements were obtained from Invitrogen (Carlsbad, CA, USA). Enhanced chemiluminescence reagent and polyvinylidene difluoride (PVDF) membrane were obtained from

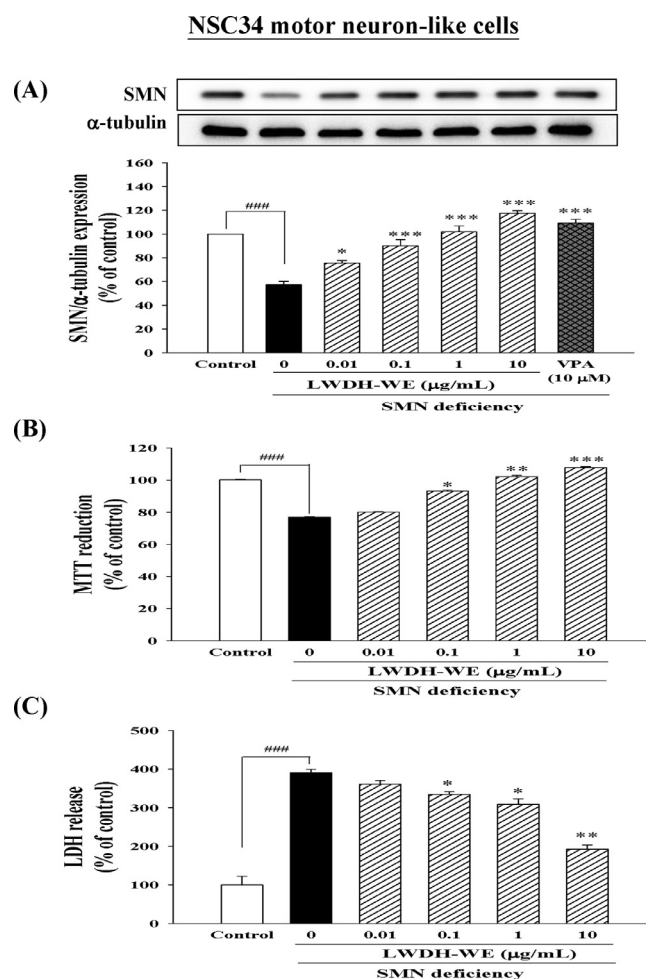


Fig. 1. Effects of LWDH-WE on protein expression of SMN (A), and neuronal cell viability assayed by MTT (B) and LDH (C) assays in SMN-deficient NSC34 cells. SMN deficiency was induced by doxycycline (1 μg/ml) treatment in SMN-specific shRNA NSC34 cells for 48 h, and cells transfected with empty vector were used as control. Cells were treated with LWDH-WE for 48 h. Valproic acid (VPA) was used as positive control drug. Densitometry analyses were presented as the relative ratio of protein/α-tubulin protein, and are represented as percentages of control group. Bars represent the mean ± SEM from three independent experiments. ### $p < 0.001$ vs. control group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. SMN-deficient group.

Millipore (Billerica, MA, USA). All materials for SDS-PAGE were obtained from Bio-Rad (Hercules, CA, USA). LDH (lactate dehydrogenase) cytotoxicity assay kit was purchased from G-Biosciences (St. Louis, MO, USA). Plasma membrane extraction kit was obtained from BioVision (Mountain View, CA, USA). Antibodies used for immunoblotting or immunostaining were as follows: β-actin (Sigma); Bcl-2, cytochrome c, RhoA, ROCK2, GAPDH, and all horseradish peroxidase-conjugated secondary antibodies (Santa Cruz, CA, USA); SMN (BD Bioscience, San Jose, CA, USA); caspase-3, p-LIMK, t-LIMK, p-cofilin, t-cofilin (Cell signaling, Danvers, MA, USA); tubulin (Millipore, Billerica, MA, USA); fluorochrome-conjugated secondary antibody (Invitrogen, Carlsbad, CA, USA).

Liuwei dihuang water extract (LWDH-WE)

Standardized LWDH-WE were obtained from a GMP pharmaceutical factory. The HPLC profile of LWDH-WE had been performed in our previous study (Tseng et al., 2014), and the content of chemical standards in LWDH-WE were as follow: 5-hydroxymethyl-2-furaldehyde 3.19 ± 0.07 mg/g, morroniside 5.25 ± 0.03 mg/g, loganin 2.37 ± 0.04 mg/g, paeoniflorin 2.12 ± 0.02 mg/g, verbascoside

NSC34 motor neuron-like cells

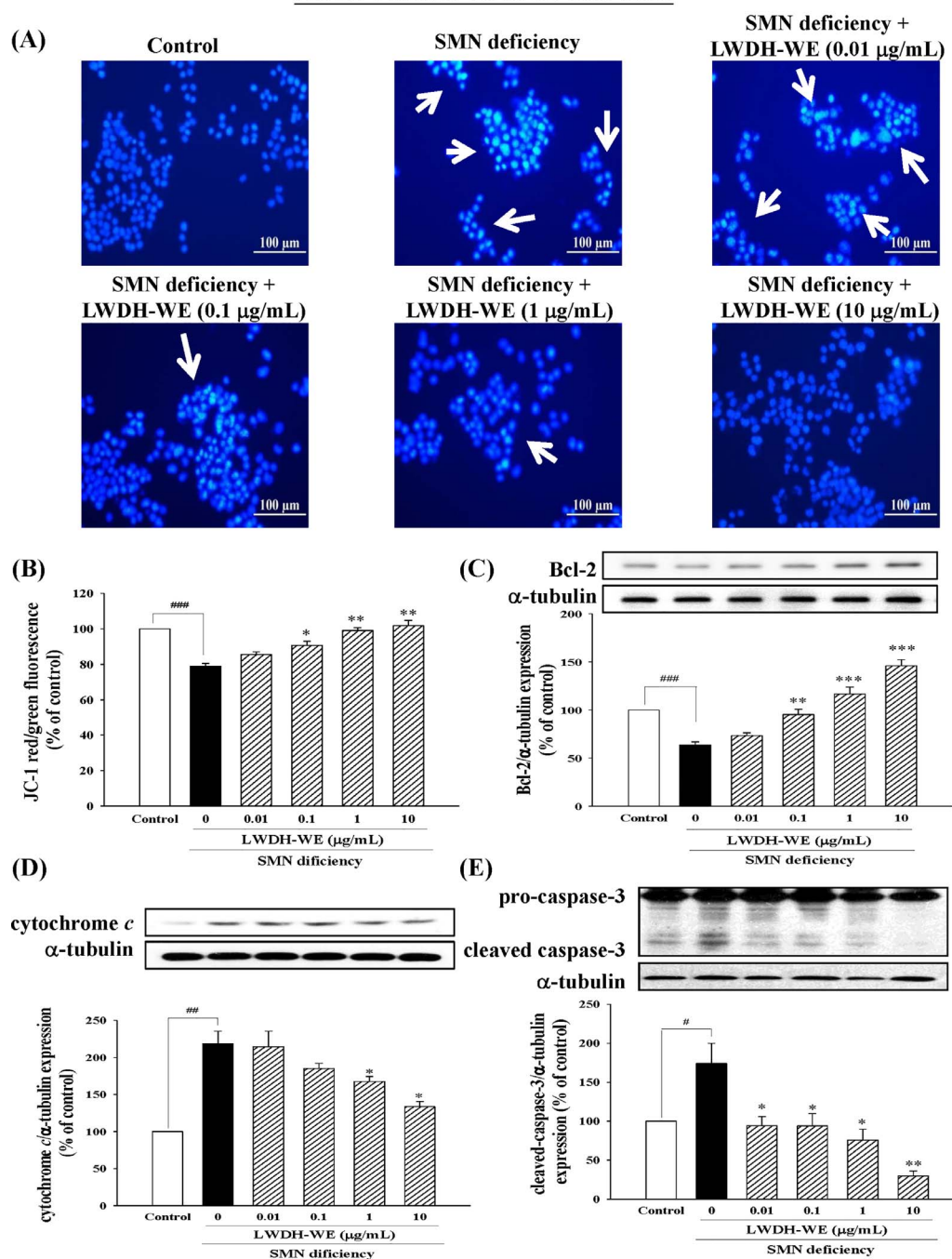


Fig. 2. Effects of LWDH-WE on nuclear condensation (A), changes of mitochondrial membrane potential ($\Delta\Psi\text{m}$) (B), and the expressions of Bcl-2 (C), cytosolic cytochrome c (D), and cleaved caspase-3 (E) in SMN-deficient NSC34 cells. SMN deficiency was induced by doxycycline (1 $\mu\text{g/mL}$) treatment in SMN-specific shRNA NSC34 cells for 48 h, and cells transfected with empty vector were used as control. Cells were treated with LWDH-WE for 48 h. Nuclear condensation (white arrow) was determined by Hoechst 33,342 and observed by a fluorescent microscope. Scale bar = 100 μm . $\Delta\Psi\text{m}$ was measured by JC-1 staining and analyzed by a flow cytometer. Densitometry analyses were presented as the relative ratio of protein/ α -tubulin protein, and are represented as percentages of control group. Bars represent the mean $\pm$ SEM from three independent experiments. # p < 0.05, ## p < 0.01, ### p < 0.001 vs. control group. * p < 0.05, ** p < 0.01 vs. SMN-deficient group.

0.10 $\pm$ 0.00 mg/g and paeonol 1.14 $\pm$ 0.01 mg/g (Tseng et al., 2014).

Cell cultures

NSC34 cells transfected with pcDNA6/TR vector or pSuperior-SMN shRNA were obtained from Dr. Hung Li at Academia Sinica (Wen et al., 2010). Human SMA fibroblast cell lines were established from SMA patients' skin biopsies with informed approval by the Institutional Review Board of Kaohsiung Medical University Chung-Ho Memorial Hospital (KMUH-IRB-990,122). Cells were cultured in DMEM supplemented with 10% FBS, 100 U/ml penicillin, 100 $\mu\text{g/mL}$ streptomycin, and 0.25 $\mu\text{g/mL}$ amphotericin B at 37 $^{\circ}\text{C}$ in a humidified atmosphere containing 5% CO_2 . NSC34 cells were then differentiated by serum-free defined medium for 6 days. SMN deficiency was induced by

doxycycline (1 $\mu\text{g/mL}$) treatment in SMN-specific shRNA NSC34 cells for 48 h according to our previous study (Tseng et al., 2016).

Animal preparation and drug treatment

The Animal Care and Use Committee at the Kaohsiung Medical University approved the animal use and methods (approval ID: 101,129). All animals used in this study were fed in the Animal Center of Kaohsiung Medical University under constant temperature and 12 h light-dark cycle. Transgenic mice expressing the severe SMA-like phenotypes, with a mean survival of 10.2 $\pm$ 0.7 days, were generated from breeder pairs obtained from The Jackson Laboratory (FVB.Cg-Tg(SMN2 $^{\Delta\text{delta}7}$)4299Ahmb Tg(SMN2)89Ahmb Smn1 $^{\text{tm}1\text{Ms}d}$, stock number: 0,05,025) (Le et al., 2005), and the genotype of mice was performed by

NSC34 motor neuron-like cells

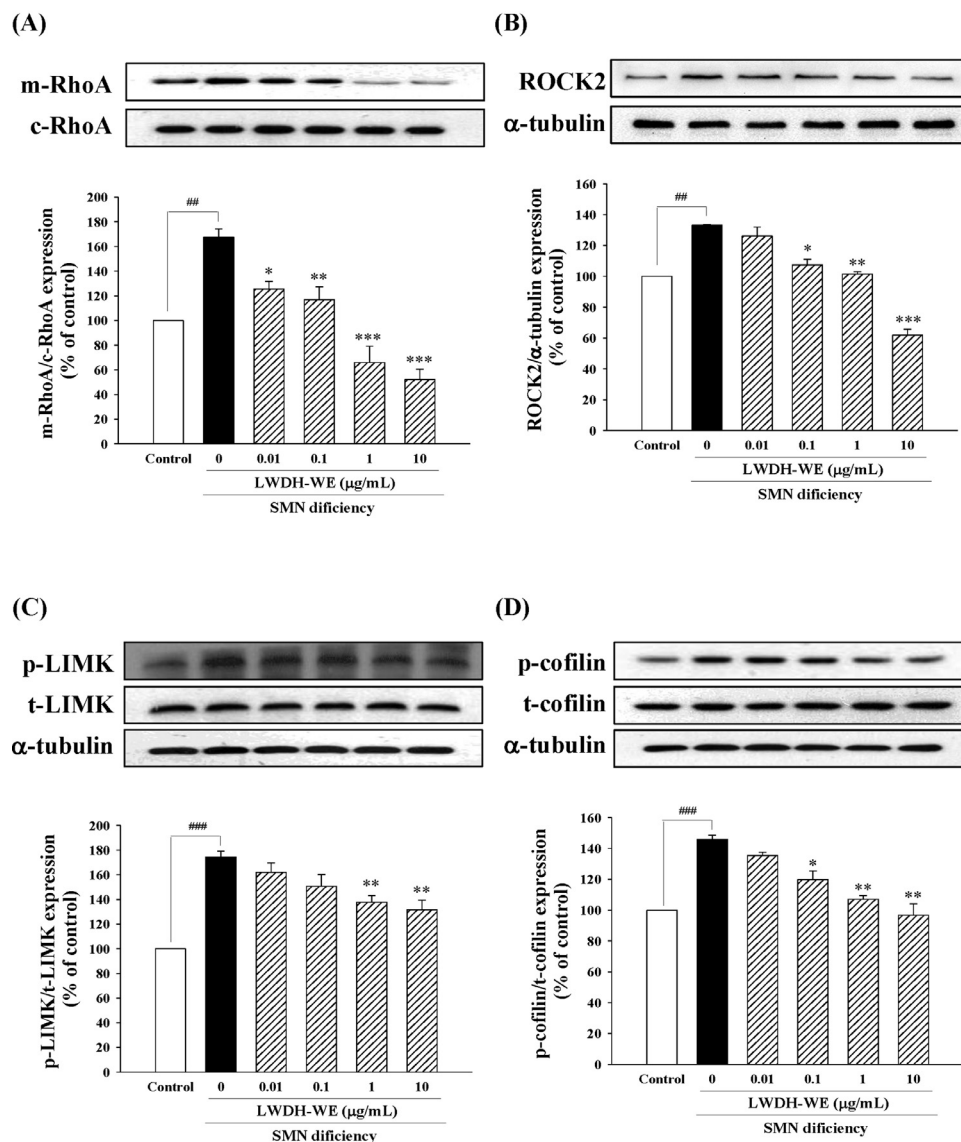


Fig. 3. Effects of LWDH-WE on protein expressions of membrane-RhoA (m-RhoA) (A), ROCK2 (B), p-LIMK (C), and p-cofilin (D) in SMN-deficient NSC34 cells. SMN deficiency was induced by doxycycline (1 μg/ml) treatment in SMN-specific shRNA NSC34 cells for 48 h, and cells transfected with empty vector were used as control. Cells were treated with LWDH-WE for 48 h. Densitometry analyses were presented as the relative ratio of membrane-protein (m-protein)/cytosolic-protein (c-protein), protein/α-tubulin protein, or p-protein/t-protein and are represented as percentages of control group. Bars represent the mean ± SEM from three independent experiments. ## $p < 0.01$, ### $p < 0.001$ vs. control group. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. SMN-deficient group.

PCR of DNA from tail biopsies (Tseng et al., 2016). Mice were subcutaneously received LWDH-WE daily or equal volume of saline from postnatal day 1 (P1). Animals were divided into wild-type (WT) group, SMA group, and two drug treatment groups (SMA mice treated with 15 and 30 mg/kg/day LWDH-WE, respectively). Daily weights were measured from P1, and survival rates were recorded. Spinal cord, brain, and gastrocnemius muscle tissues were dissected from animals of P7 for western blot analysis.

Cell viability assay

Cell viability was determined by MTT and LDH assay. In MTT assay, cells were firstly incubated with 0.5 mg/ml MTT for 3 h in 37 °C. The Formazan crystals in the cells were dissolved with 100 μl DMSO, and absorbance was read at 560 nm. In LDH assay, culture medium was collected and LDH release was measured by a cytotoxicity detection kit according to the manufacturer's guide. The tetrazolium salts produced in the LDH-induced enzymatic reaction were then reduced to red formazan. Absorbance was read at 490 nm by a microplate reader.

Immunocytochemistry

Briefly, cells were firstly fixed with 4% paraformaldehyde for 30 min. Cells were then incubated with permeabilizing solution (0.1% Triton X-100 in PBS) for 10 min, and with blocking buffer (2% BSA in PBS) for 1 h. Cells were then stained with Hoechst 33,342 for 10 min or incubated with specific primary antibodies overnight at 4 °C. Appropriate secondary antibodies Alexa Fluor 488 goat anti-rabbit IgG or Alexa Fluor 555 goat anti-mouse IgG were used respectively. Images were collected under a fluorescence microscopy (Nikon, Japan) or confocal laser scanning microscope (Zeiss, Germany). Hoechst 33,342 staining was assessed to detect the features of apoptotic cells (DNA condensation and nuclear fragmentation) and tubulin staining was assessed to observe neurites morphology. The neurite outgrowth was determined by calculating the mean neurite length using Image J software as our previous studies (Hsu et al., 2013; Tseng et al., 2016).

Measurement of mitochondrial membrane potential ($\Delta\Psi_m$)

The changes of $\Delta\Psi_m$ was examined by JC-1 fluorescence staining

NSC34 motor neuron-like cells

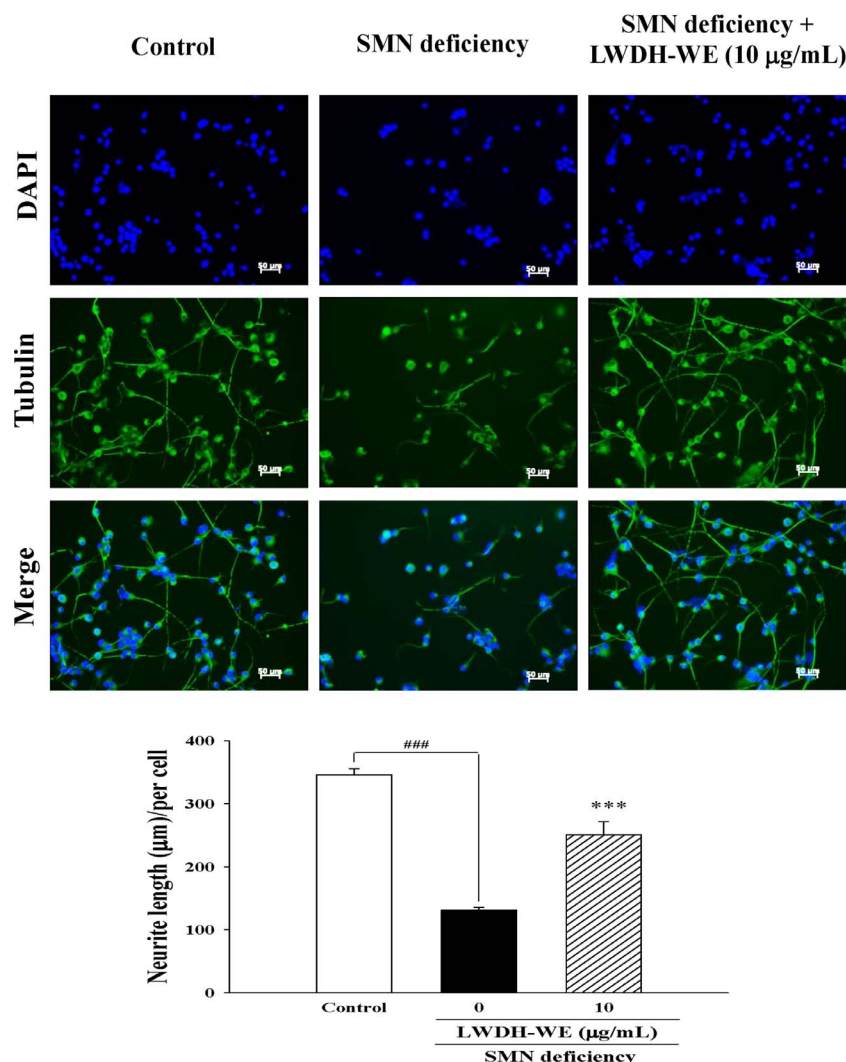


Fig. 4. Effect of LWDH-WE on neurite length in SMN-deficient NSC34 cells. SMN deficiency was induced by doxycycline (1 µg/ml) treatment in *SMN*-specific shRNA NSC34 cells for 48 h, and cells transfected with empty vector were used as control. Cells were treated with LWDH-WE for 48 h. Neurites were confirmed by fluorescent image of tubulin (green) staining under a fluorescent microscope. DAPI (blue) was used for nuclei staining. Scale bar = 50 µm. Neurite length was calculated by using Image J software. Bars represent the mean ± SEM from three independent experiments. *** $p < 0.001$ vs. control group. *** $p < 0.001$ vs. SMN-deficient group. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and analyzed by Coulter CyFlow Cytometer (Partec, Germany). JC-1 accumulates in the mitochondria as aggregates with red fluorescence of normal cells. While undergoing apoptosis, JC-1 translocates to cytosol and accrues as green fluorescence monomer. $\Delta\Psi_m$ was represented as ratio of JC-1 red fluorescence to JC-1 green fluorescence. Red JC-1 fluorescence was detected by excitation/emission at 540/570 nm, and green JC-1 fluorescence was detected by excitation/emission at 495/520 nm.

Western blot analysis

Cells and mouse tissues were homogenized or lysed by lysis buffer (Thermo Scientific, Waltham, MA, USA). Membrane protein extracts were obtained by using commercial kit followed the manufacturer guidelines. Protein concentration was determined by using protein Bio-Rad assay kit, and equal amounts of protein were separated on polyacrylamide gel and transferred to PVDF membranes. Non-specific binding was blocked with TBST (50 mM Tris-HCl, pH 7.6, 150 mM NaCl, 0.1% Tween 20) containing 5% non-fat milk for 1 h at room temperature. Each membrane was then incubated with appropriate primary and secondary antibodies, and was visualized with the enhanced chemiluminescence reagent.

Hind-limb suspension test (HLST)

HLST is a non-invasive motor function test designed for neonatal rodents to evaluate muscle strength of hind-limb. Briefly, mice were placed head down, hanging by the hind limbs in a 50 ml plastic centrifuge tube and recorded the hind-limb score (HLS). HLS was evaluated by assessing the position of legs and tail. The criteria are as follows: score 4, normal hind-limb separation with tail raised; score 3, hind limbs closer together but seldom touch each other; score 2, hind limbs close to each other and often touch; score 1, hind limbs always appeared in a clasped position with tail raised; score 0, constant clasping of the hind limbs with tail lowered or failure to hold onto the tube.

Statistical analysis

Statistical analysis was performed using InStat version 3.0 (GraphPad Software, San Diego, CA, USA). Survival was analyzed by Kaplan–Meier curves and log-rank test followed by Bonferroni *post hoc* test for multiple comparisons. Data were expressed as means ± SEM, and statistical significance was analyzed using one-way ANOVA followed by Dunnett's test for all pair comparisons. Significance was set at a *P*-value of less than 0.05.

Skin fibroblasts from type I SMA patient

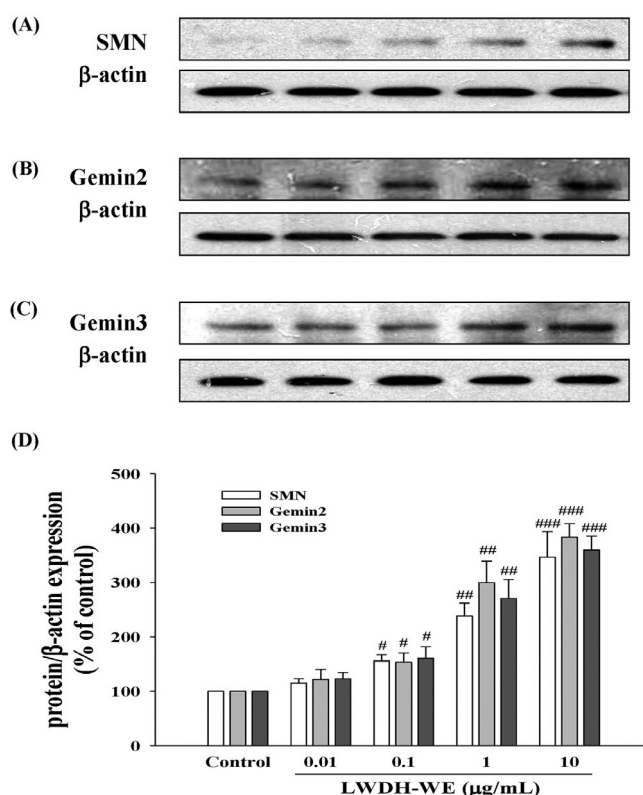


Fig. 5. Effects of LWDH-WE on protein expressions of SMN (A), Gemin2 (B), and Gemin3 (C) in skin fibroblasts from type I SMA patient. Cells were treated with LWDH-WE (0.01–10 μg/ml) for 48 h (D) Densitometry analyses are presented as the relative ratio of protein/β-actin protein, and are represented as percentages of control group. Data represented as mean ± S.E.M. # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$ vs. control.

Results

Effects of LWDH-WE on SMN protein level and cell viability of SMN-deficient NSC34 cells

Because SMN deficiency is one of the causes of motor neuronal loss, we firstly examined the effects of LWDH-WE on the protein level of SMN. Results indicated LWDH-WE significantly increased SMN protein level in SMN-deficient NSC34 cells (Fig. 1A). We next examined the effects of LWDH-WE on cell viability of SMN-deficient NSC34 cells by MTT and LDH assays. Results indicated that LWDH-WE increased MTT reduction (Fig. 1B) and decreased LDH release (Fig. 1C), suggesting LWDH-WE increased cell viability of NSC34 cells under SMN-deficient condition.

LWDH-WE inhibits apoptotic-related pathway and decreased apoptotic death in SMN deficient- NSC34 cells

To further ascertain the effects of LWDH-WE on SMN deficiency-induced apoptotic death, Hoechst 33,342 staining was performed to detect the features of apoptotic cells (nuclear condensation) in SMN deficient-NSC34 cells. Results indicated SMN deficiency induced nuclear condensation in NSC34 cells (white arrow). LWDH-WE could decrease SMN deficiency-induced nuclear condensation (Fig. 2A), suggesting LWDH-WE attenuated SMN deficiency-induced apoptosis. Moreover, SMN deficiency induced change of mitochondrial membrane potential ($\Delta\Psi_m$), and LWDH-WE could improve mitochondrial function by increasing $\Delta\Psi_m$ in SMN-deficient NSC34 cells (Fig. 2B). Furthermore, LWDH-WE enhanced anti-apoptotic effect via up-regulation of Bcl-2 expression (Fig. 2C) and attenuated apoptotic effect via down-regulation of cytosolic cytochrome c (Fig. 2D) and cleaved caspase-3 (Fig. 2E) expressions in SMN-deficient NSC34 cells.

LWDH-WE inhibits m-RhoA/ROCK2/p-LIMK/p-cofilin pathway and increases neurite length in NSC34 cells under SMN-deficient condition

SMN protein and RhoA pathway play important roles in neurite outgrowth. Therefore, we next investigated the effects of LWDH-WE on

Skin fibroblasts from Type I SMA patients

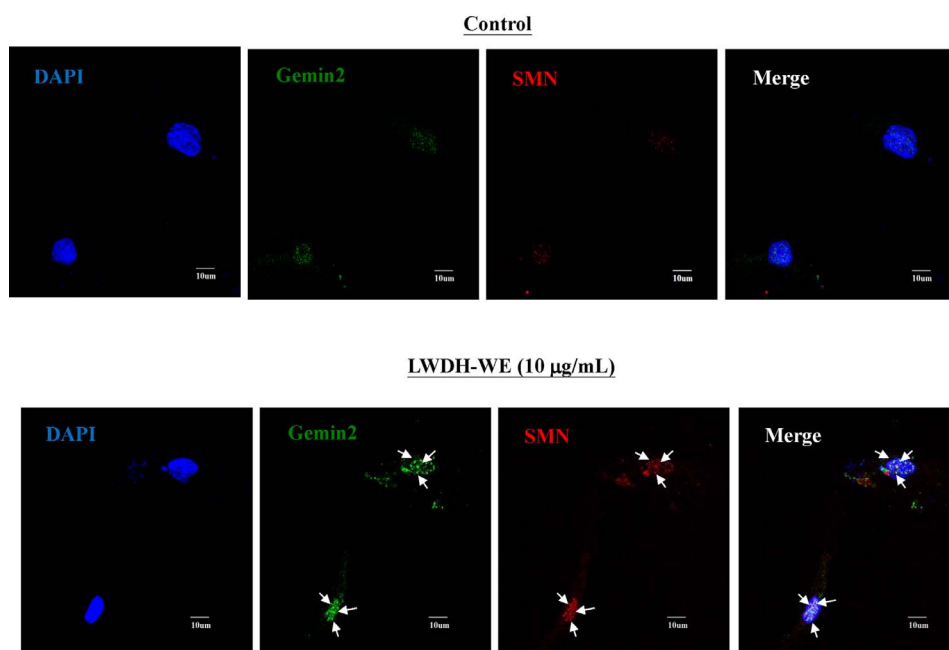


Fig. 6. Effect of LWDH-WE on SMN-containing nuclear gems in skin fibroblasts for type I SMA patient. Cells were treated with LWDH-WE for 48 h SMN-containing nuclear gems were observed by confocal microscopy with SMN (red) /Gemin2 (green) staining, and pointed by white arrow. DAPI (blue) was used for nuclei staining. Scale bar = 10 μm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

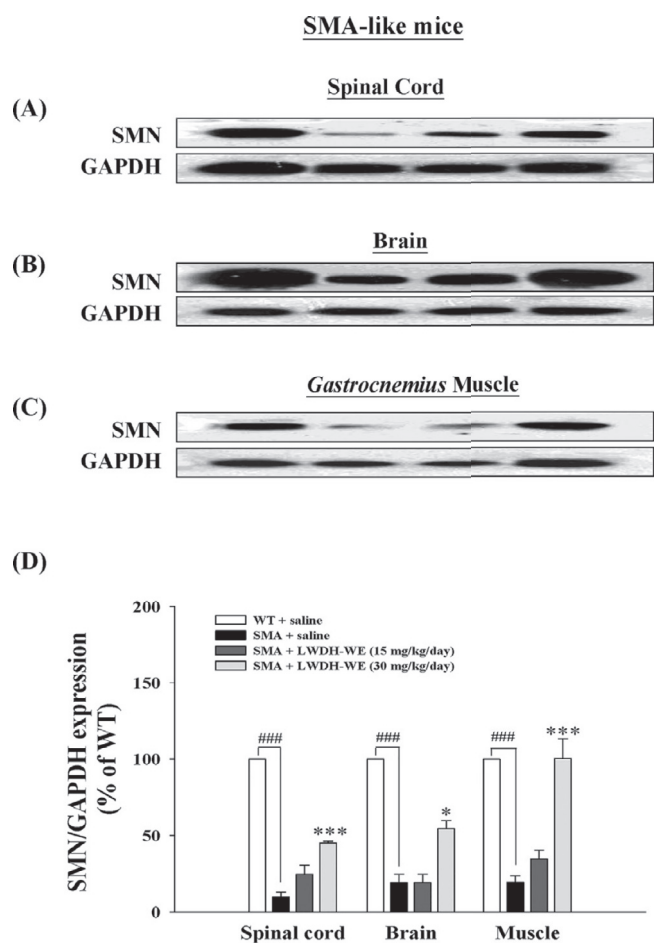


Fig. 7. Effects of LWDH-WE on protein level of SMN in spinal cord (A), brain (B), and gastrocnemius muscle (C) tissues of SMA-like mice. Mice were subcutaneously injected with LWDH-WE (15 mg/kg/day, $n = 3$ and 30 mg/kg/day, $n = 3$, respectively) or an equal amount of saline ($n = 3$), and wild-type mice were subcutaneously injected with the same amount saline as control ($n = 3$). (D) Densitometry analyses were presented as the relative ratio of protein/GAPDH protein, and were represented as percentages of the wild type (WT) group. Bars represent the mean $\pm$ SEM. **** $p < 0.001$ vs. wild type (WT) group. * $p < 0.05$, ** $p < 0.01$ vs. SMA-like mice treated with saline group.

protein expressions of RhoA and its downstream effectors including ROCK2, LIMK, and cofilin in SMN-deficient NSC34 cells. Results indicated that SMN deficiency up-regulated the expression of membrane-RhoA (m-RhoA) (Fig. 3A), the active GTP-bound form of RhoA, which induced neurite restriction. However, LWDH-WE attenuated m-RhoA expression (Fig. 3A) and RhoA downstream effector ROCK2 (Fig. 3B) induced by SMN-deficiency. Because activation of RhoA/ROCK increased phosphorylation of LIMK and cofilin, SMN deficiency enhanced phosphorylation of LIMK (Fig. 3C) and cofilin (Fig. 3D) were observed. And LWDH-WE attenuated SMN deficiency-induced increase of p-LIMK and p-cofilin (Fig. 3C and D). Since inhibition of RhoA/ROCK pathway can improve neurite outgrowth, we further confirmed the effects of LWDH-WE on neurite length by immunofluorescence staining. The result indicated that LWDH-WE significantly attenuated SMN deficiency-induced decrease of neurite length (Fig. 4).

LWDH-WE up-regulates protein levels of SMN, Gemin2, Gemin3, and increases the numbers of SMN-containing nuclear gems in human SMA fibroblasts

We further investigated the effects of LWDH-WE on SMN and SMN complex components Geminins in skin fibroblasts from type I SMA patient. Results indicated LWDH-WE significantly increased expressions of SMN (Fig. 5A and D), Gemin2 (Fig. 5B and D), and Gemin3 (Fig. 5C and

D) in skin fibroblasts from type I SMA patient. Moreover, the effect of LWDH-WE on functional SMN complex production was also examined by measuring SMN-containing nuclear gems using confocal microscope. As the result shown, LWDH-WE increased the numbers of SMN-containing nuclear gems in comparison with the control group of type I SMA skin fibroblasts (Fig. 6).

Effects of LWDH-WE on SMN protein level of neuronal and muscular tissues, and lifespan of SMA-like mice

SMA is the most identified MND related to SMN deficiency; we therefore investigated the effects of LWDH-WE on SMA using a severe SMA mouse model, SMA $\Delta 7$. Mice subcutaneously received LWDH-WE (15 and 30 mg/kg/day, respectively) or equal volume of saline from postnatal day 1. Results demonstrated that LWDH-WE could up-regulate protein levels of SMN in the spinal cord (Fig. 7A and D), brain (Fig. 7B and D), and gastrocnemius muscle (Fig. 7C and D) of SMA-like mice. Since muscle weakness and atrophy were the most prominent clinical features of SMA, we also evaluated the effects of LWDH-WE on muscle strength and body weight of SMA-like mice. Muscle strength of mice was examined by hind-limb suspension test (HLST) and hind-limb scores (HLS) were recorded. Results from HLST showed that all the mice in WT group scored 4 (Fig. 8A), representing normal hind-limbs separation and tail raise. The HLS of SMA-like mice ranged from score 0 to 1 (average score was 0.25 ± 0.16), representing clasping hind-limbs with tail lower (Fig. 8A). LWDH-WE (30 mg/kg/day)-treated SMA-like mice showed closer hind-limbs with tail raised, and the HLS was improved to 2.13 ± 0.35 on average (Fig. 8A). Moreover, LWDH-WE (30 mg/kg/day) treatment improved the body weight of SMA-like mice (1.95 ± 0.08 g) compared with SMA saline-treated mice (1.63 ± 0.07 g) from P3, and this became more obvious in the following days (Fig. 8B). However, while 15 mg/kg/day of LWDH-WE treatment also improved body weight (1.93 ± 0.08 g) of SMA-like mice from P3 (Fig. 8B), its improvement on HLS was not significant (0.88 ± 0.23 on average) (Fig. 8A). Furthermore, the lifespan of the 15 and 30 mg/kg/day of LWDH-WE-treated SMA-like mice were 12.63 ± 0.18 days and 15.13 ± 0.35 days respectively, while saline-treated SMA-like mice represented 10.00 ± 0.68 days of lifespan (Fig. 8C).

Discussion

LWDH is a widely used herbal formula prescribed as therapy or adjuvant therapy against many types of disorders for its broad spectrum of pharmacological effects and mechanisms (Zhou et al., 2016). Previously, we reported that LWDH-WE possesses neuroprotective effects against Parkinson's toxin (Tseng et al., 2014); this study further revealed its protection on SMN-deficient motor neurons, and its novel effects on improving muscle strength and body weight of SMA-like mice, providing the evidence and potential of LWDH on the application in the NMD-related neurodegeneration and muscle atrophy.

SMN is a known housekeeping protein, critical for the correct assembly of the snRNP complexes required for RNA splicing, which plays crucial roles in motor neuron physiology including development, synaptogenesis and survival (Cauchi, 2014; Talbot and Davies, 2008). Several studies have demonstrated that motor neurons are particularly vulnerable to low level of SMN, suggesting SMN enhancement is universally beneficial in MNDs such as SMA and amyotrophic lateral sclerosis (ALS) (Cauchi, 2014; Perera et al., 2016). The present study indicated that LWDH-WE is able to increase protein level of SMN in SMN-deficient NSC34 cells, skin fibroblasts of SMA patient, and neuronal and muscular tissues of SMA-like mice. Besides, it is important for SMN to form complex with Geminins for sustaining stability. Results from SMA fibroblasts also demonstrated LWDH-WE could up-regulated the expressions of Geminins and SMN-containing nuclear gems, suggesting LWDH-WE could improve SMN stability. LWDH-WE also attenuated

SMA-like mice

(A)

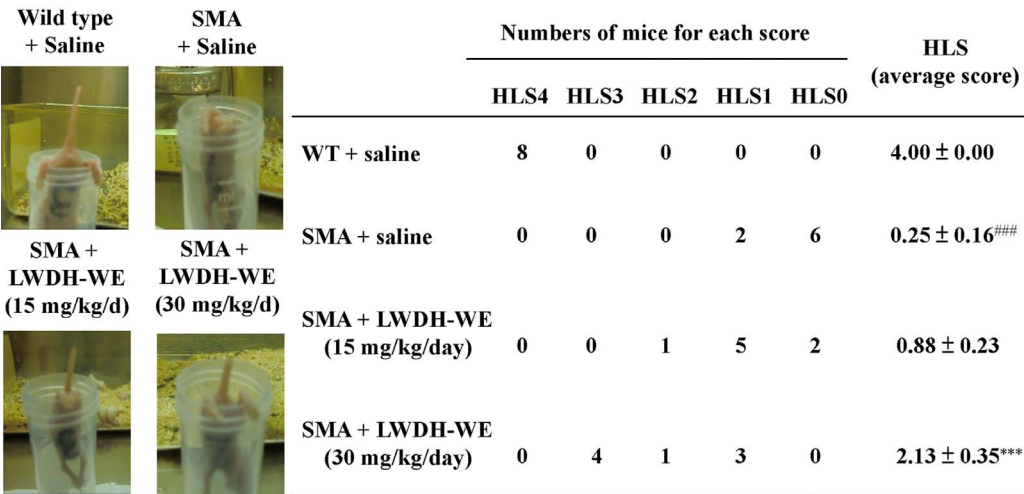
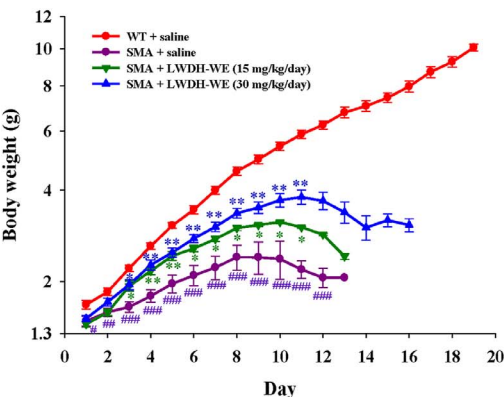
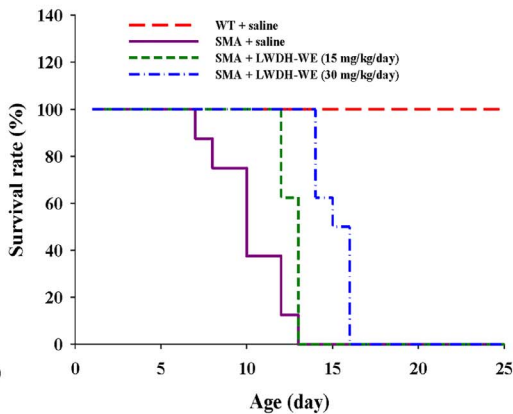


Fig. 8. Effects of LWDH-WE on muscle strength (A), body weight (B), and survival (C) of SMA-like mice. SMA mice were subcutaneously injected with LWDH-WE (15 mg/kg/day, n = 8 and 30 mg/kg/day, n = 8, respectively), or an equal amount of saline (n = 8), and wild-type mice were subcutaneously injected with the same amount saline as control (n = 8). Muscle strength was assayed by hind-limb score (HLS) during hind-limb suspension test (HLST). HLS was evaluated by assessing the position of legs and tail, and the criteria are as follows: score 4, normal hind-limb separation with tail raised; score 3, hind limbs closer together but seldom touch each other; score 2, hind limbs close to each other and often touch; score 1, hind limbs always appeared in a clasped position with tail raised; score of 0, constant clasping of the hind limbs with tail lowered or failure to hold onto the tube. Kaplan–Meier survival curves of WT mice and SMA-like mice treated with LWDH-WE or saline. $p < 0.05$, log-rank test. Bars represent the mean $\pm$ SEM. [#] $p < 0.05$, ^{##} $p < 0.01$, ^{###} $p < 0.001$ vs. wild type (WT) group. ^{*} $p < 0.05$, ^{**} $p < 0.01$, ^{***} $p < 0.001$ vs. SMA mice treated with saline group.

(B)



(C)



SMN deficiency-induced motor neuronal death in NSC34 cells. Besides, SMN deficiency is known to cause motor neuronal degeneration through apoptotic mechanism (Gallotta et al., 2016; Parker et al., 2008), and can impair mitochondrial activity of motor neurons (Miller et al., 2016). B cell lymphoma 2 (Bcl-2) family members act on key step of mitochondrial apoptotic pathway via regulating mitochondrial outer membrane permeabilization. Bcl-2 is one of the anti-apoptotic members of Bcl-2 family, which can stabilize mitochondrial function, and synergize with SMN leading to increase of the anti-apoptotic ability and decrease of pro-apoptotic protein-induced apoptosis (Anderton et al., 2013). The present results indicated that LWDH-WE increased mitochondrial membrane potential ($\Delta\Psi_m$) and Bcl-2 expression in SMN-deficient cells. Moreover, while apoptosis occur, cytochrome c redistributes from mitochondria to cytosol and further cause caspase-3 cleavage (Wu and Bratton, 2013). The present results also indicated that LWDH-WE attenuated SMN deficiency-induced apoptotic death via decrease of nuclear condensation, cytosolic cytochrome c and cleaved caspase3. The above evidence validates the beneficial effects of LWDH-WE on SMA for its SMN-promoting and anti-apoptotic activities.

Over-activation of the RhoA/ROCK pathway has been found in different mouse models of MNDs including SMA (Bowerman et al., 2010) and ALS (Takata et al., 2013). Therefore, inhibition of the RhoA/ROCK pathway is suggested as a viable therapeutic target for MNDs (Bowerman et al., 2010; Chong et al., 2017; Coque et al., 2014; Takata et al., 2013). RhoA/ROCK pathway is an important negative regulating pathway of neurite outgrowth, which leads to growth cone collapse and neurite retraction (Gallo and Letourneau, 2004). The present results demonstrated that LWDH-WE attenuated SMN deficiency-induced neurite retraction. LWDH-WE also attenuated RhoA membrane expression and ROCK2 expression in SMN-deficient NSC34 cells. Moreover, activation of RhoA/ROCK can lead to LIM kinase (LIMK) activation, which then inactivates cofilin by phosphorylation and causes actin cytoskeleton rearrangement. LWDH-WE also attenuated SMN deficiency-induced increase of p-LIMK and p-cofilin in NSC34 cells. Taken together, in addition to SMN promotion and anti-apoptotic mechanisms, RhoA/ROCK inhibition is therefore a noteworthy mechanism of LWDH-WE for neuroprotection.

Because MNDs can ultimately cause muscle atrophy and weakness (Rossi et al., 2012; Saxena and Caroni, 2011), targeting on

neuroprotection with enhancement of muscle strength has become an important issue for drug development on MND therapeutics. Fasudil, a well-known RhoA/ROCK inhibitor, has been found to improve survival and promotes skeletal muscle development in ALS (Takata et al., 2013). Another RhoA/ROCK inhibitor, Y27632, is also found to prolong survival and increase muscle fiber size in SMA mice (Bowerman et al., 2010). Our results revealed that LWDH-WE possesses RhoA/ROCK inhibitory activity in SMN-deficient NSC34 motor neuronal cells.

We therefore considered the possibility of LWDH-WE on muscle enhancement in SMA. Since SMAΔ7 mice are noticeably smaller in size and show signs of muscle weakness, we measured the body weight and muscle strength of disease-induced mice to evaluate the improvement of LWDH-WE on the disease symptoms of SMA-like mice. The present results demonstrated that LWDH-WE attenuated weight loss of SMA-like mice and improved HLS of SMA-like mice from 0.25 ± 0.16 to 2.13 ± 0.35 in average. Though the improvement of LWDH-WE on lifespan of SMA-like mice was not very attractive, it should be noticed that our in vivo results revealed that LWDH-WE could provide symptomatic relief of disease-induced mice by improving muscle strength and body weight.

Conclusions

LWDH-WE protects motor neurons against SMN deficiency via targeting SMN promotion, anti-apoptosis, and RhoA/ROCK inhibition. The present results provide evidence that LWDH-WE, the well-known traditional Chinese medicine, possesses multi-protective effects on neurodegeneration and muscle atrophy in experimental models of SMA, suggesting the potential of LWDH-WE as a complementary drug for SMA. The myogenic benefits of LWDH will be further studied in the future.

Conflict of interest

The authors declare no conflict of interests.

Acknowledgments

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